T.C.

Özyeğin University

CS.423-

Homework Assignment 1

2025-2026 Spring Semester

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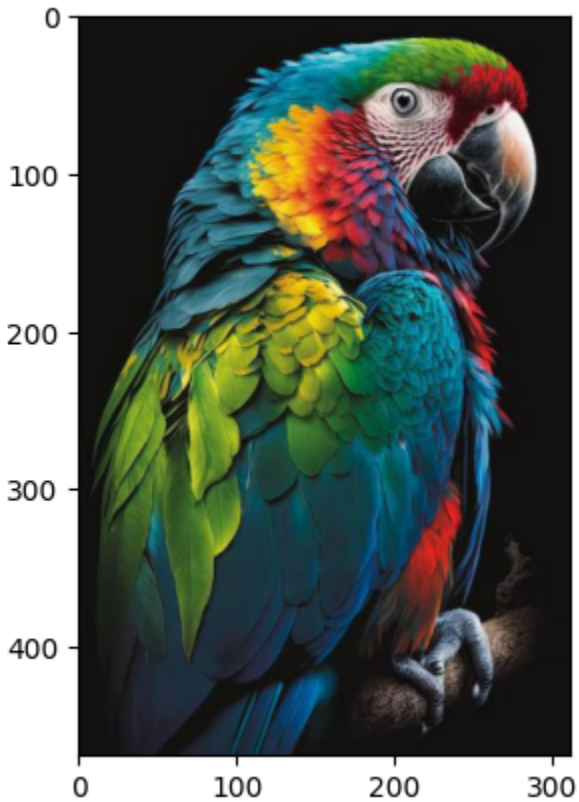
1. Before starting the assignment, drive mounted and needed libraries such as cv2 are imported. The .jpg file that will be used for this assignment is found and loaded from the CS423 folder.

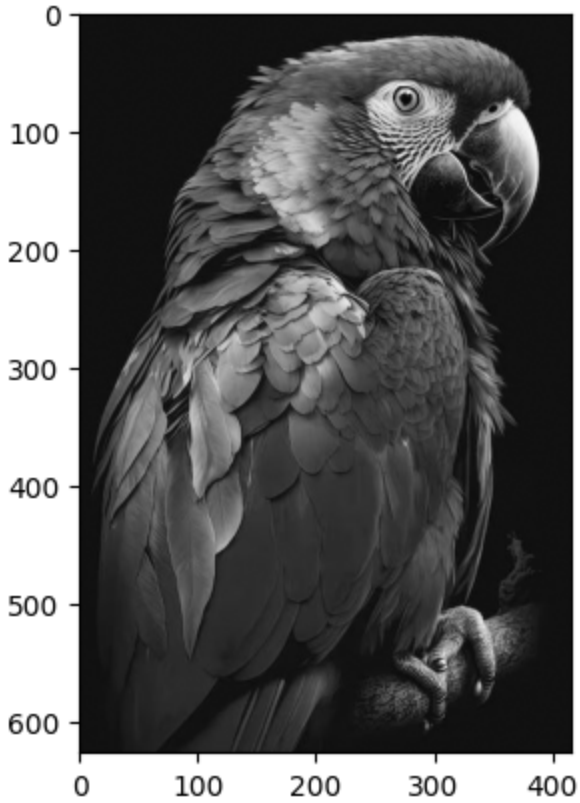
   a) When scaling the image, the “scaling factor” method is used. Width and height of the original image is multiplied by the scaling factor of 0.75, and image is resized with respect to the multiplied sizes. Resized image displayed with colors converted from BGR to RGB. Copy of the resized image is downloaded as png and the image is read to prove saving is successfully completed.

   Output: <matplotlib.image.AxesImage at 0x7b0fa93c7f50>



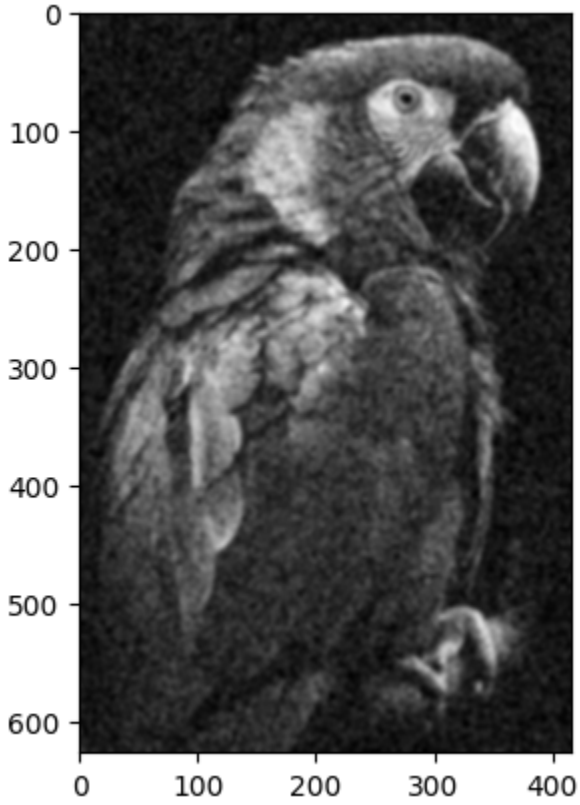
   b) The original image is copied and colors are converted from BGR to gray. When displaying the image *cmap = ’gray’* is used to correctly convert.

   Output: <matplotlib.image.AxesImage at 0x7b0f8e701e50>

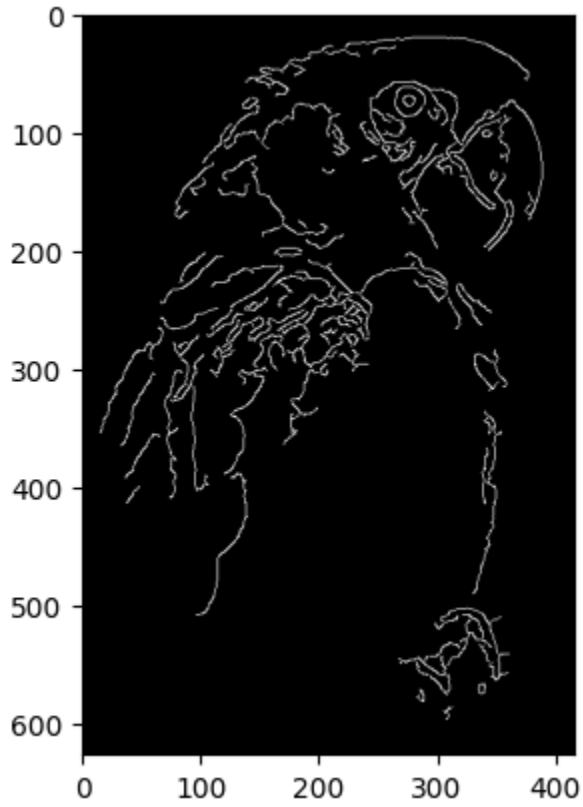


Before adding the Gaussian blur, noise is added to the gray image. After noise, gaussian blur function is used when sigma is 2.

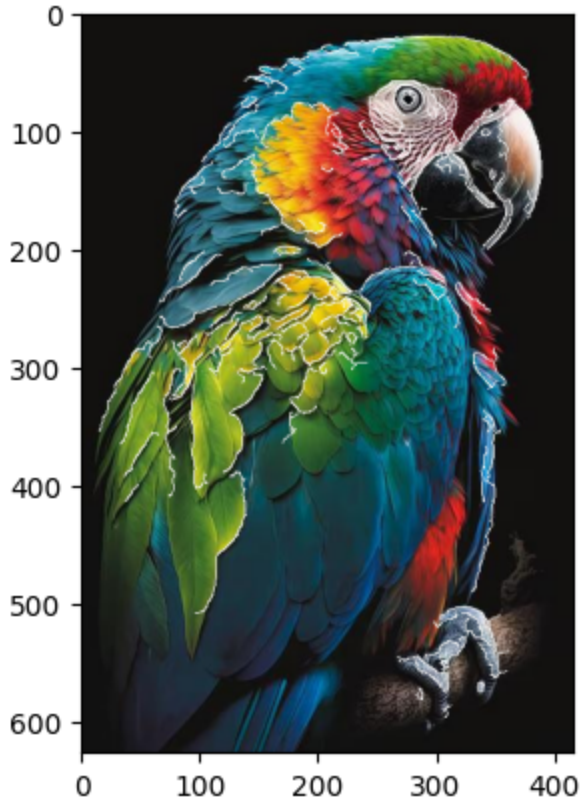
Output: <matplotlib.image.AxesImage at 0x7b0f8e2de870>



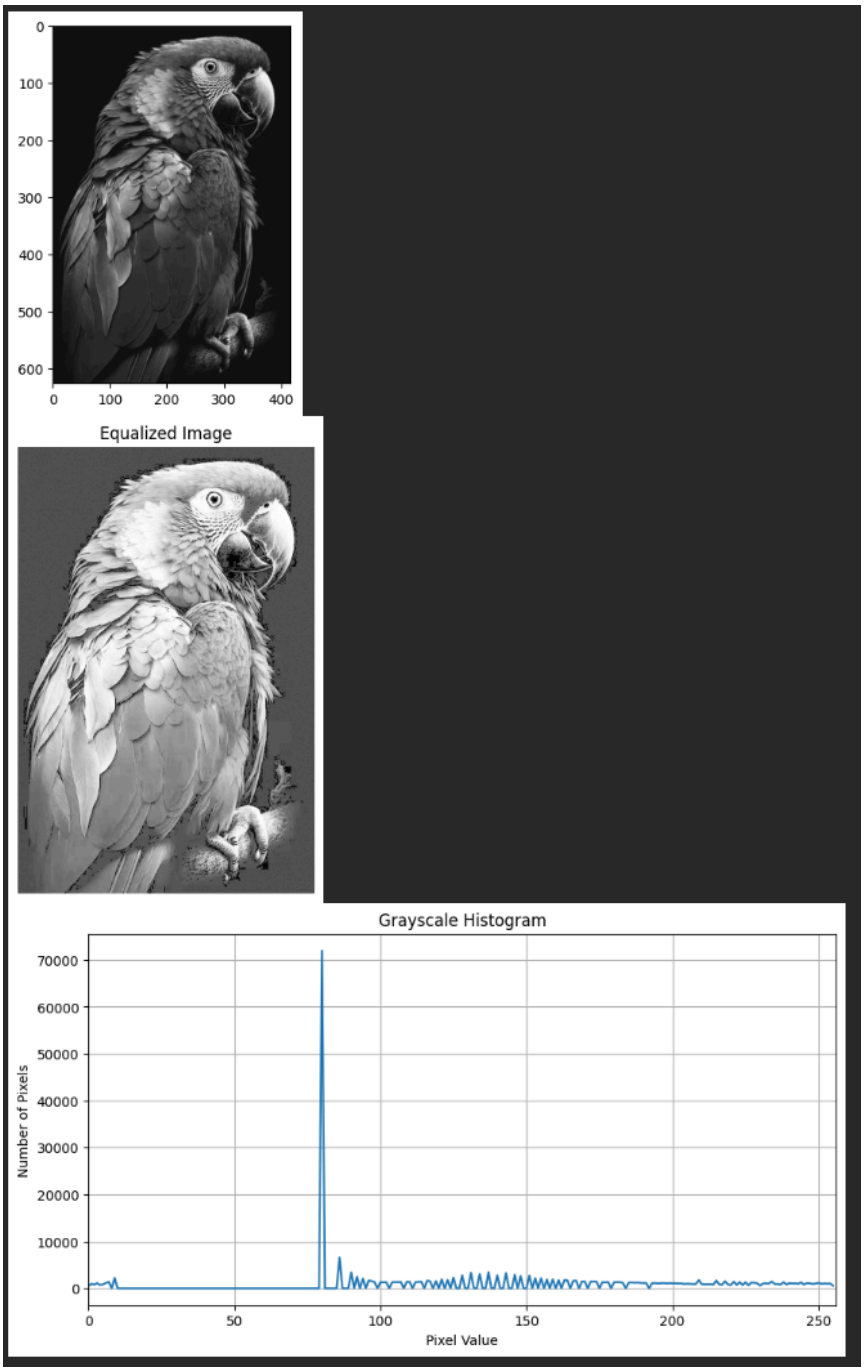
c) Blurred image from before is copied and used in the Canny function
Output: <matplotlib.image.AxesImage at 0x7b0f8dd38f20>



d) For masking, binary and method was used but after carefully reading the instructions, the cv2.addWeighted function was used because it was requested.
Output: <matplotlib.image.AxesImage at 0x7b0f8de551c0>



e) Apart from the tutorial, the “cv2.equalizeHist” function was used. After that, the data to be used in the graph was calculated using the “cv2.calcHist” function.
Output:



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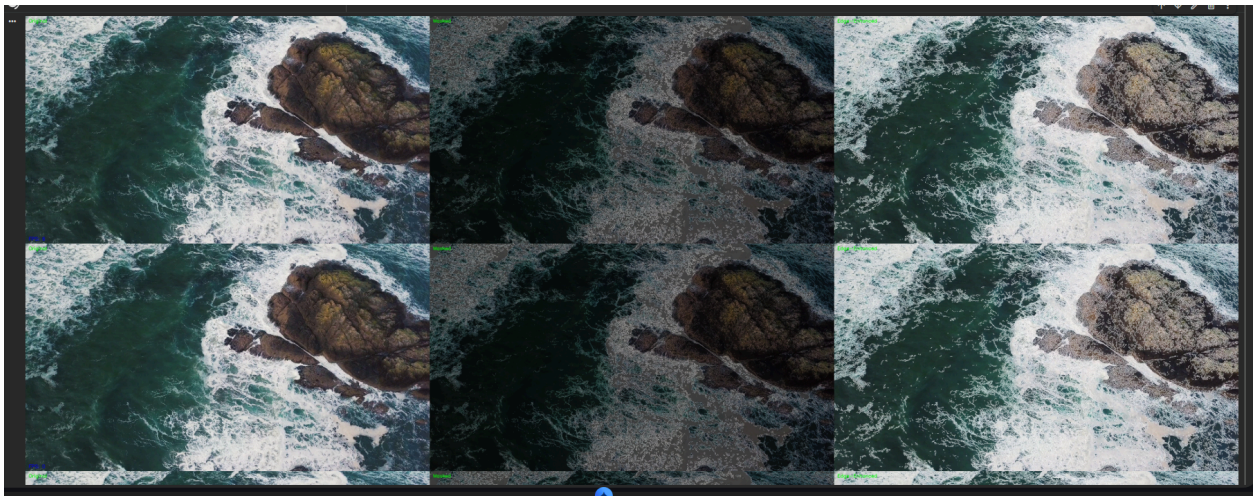
2. Before starting the assignment, drive mounted and needed libraries such as cv2 are imported. Apart from the first question, time is imported to calculate the fps. The .mp4 file that will be used for this assignment is found and loaded from the CS423 folder. Captured video is opened.

- For the parts a, b, c, d and e; functions learned from the first part are used.
- For f, np.hstack function is used to concatenate the required frames.
- For g, the difference between previous and current time is taken to calculate the fps. For taking the time, “time” is imported at the beginning of the assignment.

- Unfortunately, interrupting the stream with keyboard “q” failed. Although KeyboardInterrupt is used correctly and q is pressed, without stopping the run from the Colab stream does not stop. cv2.destroyAllWindows() and clear_output(wait=True) is used to clear buffer.

  Output: